

1

PRIOR STATE

state belief $b_{t-1}(s)$

Emotion disclosure 0.30

Emotion processing 0.45

Considering solutions 0.25

2

PREDICT

transition model
 $P(s_t | s_{t-1})$ 

$$\hat{b}_t(s_t) = \sum_{s_{t-1} \in S} P(s_t | s_{t-1}) \cdot b_{t-1}(s_{t-1})$$

predicted distribution $\hat{b}_t(s_t)$

3

LABEL RESPONSE

"That sounds
really difficult."candidate
response r_t

LLM

observed label o_t
Empathy / validation

predicted state distribution

response-derived observation

4

BAYESIAN UPDATE

observation o_t prediction $\hat{b}_t$

$$b_t(s_t) = \frac{P(o_t | s_t) \times \hat{b}_t(s_t)}{\sum_{s' \in S} P(o_t | s') \hat{b}_t(s')}$$

posterior state belief $b_t(s)$

Disclosure 0.12

Processing 0.66

Solutions 0.22

5

STYLE SCORE

$$\text{Score}(r_t) = \sum_{s \in S_{\text{target}}} b_t(s)$$

 S_{target}
Processing + Solutions

0.66 + 0.22

0.88

style compatibility